

Transport of source matrices

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Chapter 1

Transport of source matrices

Definition 1.1 (Sampled Schmidt-source operators). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) Let A, C be finite sets with nonnegative weights λ_a, μ_c , and let k be a positive integer. In the paired coordinate bases, put $|\eta\rangle = \sum_a \sqrt{\lambda_a} |a, a\rangle$ and $|\tilde{\eta}\rangle = \sum_c \sqrt{\mu_c} |c, c\rangle$. For independent standard circular complex Gaussian coordinates $g_{ac}^{(j)}$, with $\mathbb{E}|g_{ac}^{(j)}|^2 = 1$, set

$$(U_j)_{ac} = (\lambda_a \mu_c)^{1/4} g_{ac}^{(j)}, \quad (1.1)$$

$$(V_j)_{bd} = (\lambda_b \mu_d)^{1/4} \overline{g_{bd}^{(j)}}, \quad (1.2)$$

$$\widehat{X} = \frac{1}{k} \sum_{j=0}^{k-1} U_j \otimes V_j, \quad (1.3)$$

$$X = |\eta\rangle\langle\tilde{\eta}|, \quad (1.4)$$

$$\Delta = \widehat{X} - X. \quad (1.5)$$

Thus $X_{(a,b),(c,d)} = \sqrt{\lambda_a \mu_c} \delta_{ab} \delta_{cd}$, and

$$\widehat{X}_{(a,b),(c,d)} = (\lambda_a \lambda_b \mu_c \mu_d)^{1/4} \frac{1}{k} \sum_{j=0}^{k-1} g_{ac}^{(j)} \overline{g_{bd}^{(j)}}. \quad (1.6)$$

This is [Ope26, Equation (5.6)].

Definition 1.2 (Transport through four endpoint maps). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) Let A, C, W, X, Y, Z be finite index sets. Choose arbitrary fixed endpoint maps $E : \mathbb{C}^A \rightarrow \mathbb{C}^W$, $F : \mathbb{C}^A \rightarrow \mathbb{C}^X$, $\tilde{E} : \mathbb{C}^C \rightarrow \mathbb{C}^Y$, and $\tilde{F} : \mathbb{C}^C \rightarrow \mathbb{C}^Z$. For nonnegative real weights λ, μ , define

$$|\eta_{E,F}\rangle = \sum_a \sqrt{\lambda_a} E|a\rangle \otimes F|a\rangle, \quad (1.7)$$

$$X_{\text{amb}} = |\eta_{E,F}\rangle\langle\tilde{\eta}_{\tilde{E},\tilde{F}}|, \quad (1.8)$$

$$\mathcal{T}(M) = (E \otimes F)M(\tilde{E} \otimes \tilde{F})^\dagger. \quad (1.9)$$

The second ambient vector uses the weights μ . For the coordinate sampled factors of Defini-

tion 1.1, put $U_j^{\text{amb}} = EU_j\widetilde{E}^\dagger$ and $V_j^{\text{amb}} = FV_j\widetilde{F}^\dagger$. Then

$$\mathcal{T}(U \otimes V) = (EU\widetilde{E}^\dagger) \otimes (FV\widetilde{F}^\dagger), \quad (1.10)$$

$$\widehat{X}_{\text{amb}} = \frac{1}{k} \sum_{j=0}^{k-1} U_j^{\text{amb}} \otimes V_j^{\text{amb}} = \mathcal{T}(\widehat{X}), \quad (1.11)$$

$$X_{\text{amb}} = \mathcal{T}(X), \quad (1.12)$$

$$\Delta_{\text{amb}} = \widehat{X}_{\text{amb}} - X_{\text{amb}} = \mathcal{T}(\Delta). \quad (1.13)$$

In particular, the represented ambient vector is the endpoint transport of its paired coordinate Schmidt vector. These are the ambient-coordinate forms of [Ope26, Equation (5.6)].

Theorem 1.3 (Rank-one expansion of transported sources). [Lean] [✓ Stmt] *Let A, C be finite index sets and let W, X, Y, Z be arbitrary index sets. For endpoint matrices $E, F, \widetilde{E}, \widetilde{F}$ of the indicated sizes, define the coordinate arrays $u_{ab}(w, x) = E_{wa}F_{xb}$ and $v_{cd}(y, z) = \widetilde{E}_{yc}\widetilde{F}_{zd}$. Every matrix M with rows indexed by $A \times A$ and columns indexed by $C \times C$ satisfies*

$$\mathcal{T}(M) = \sum_{a,b \in A} \sum_{c,d \in C} M_{(a,b),(c,d)} u_{ab} v_{cd}^\dagger. \quad (1.14)$$

No orthonormality or spanning condition on the endpoint matrices is needed.

Proof. [✓ Proof] Expand both finite matrix products in $\mathcal{T}(M) = (E \otimes F)M(\widetilde{E} \otimes \widetilde{F})^\dagger$. The term indexed by (a, b, c, d) at row (w, x) and column (y, z) is $M_{(a,b),(c,d)} E_{wa} F_{xb} \widetilde{E}_{yc} \widetilde{F}_{zd}$, which is the corresponding entry of the rank-one summand in (1.14). $\square$

Bibliography

- [Ope26] OpenAI. Polynomial peps approximation of gapped square-grid ground states, 2026. September 24, 2026; source revision `adc7f1241b42e322a6451854ab7e4b4c146bf78a`.